

Vincent Tong

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RELEVANT EXPERIENCE

[Lions Eye Institute](#)

Perth, Australia

Software Engineering Intern

July 2025 - October 2025

- Built a cross-platform medical application (C++/Qt, Python) with a SQLite data layer, **centralising ophthalmic data** and **reducing inspection time by 89%**.
- Developed a high-performance image processing pipeline utilizing a **pretrained model**; engineered modular preprocessing and evaluation scripts that **achieved 83% accuracy** for **retinal vessel segmentation** across heterogeneous datasets.
- Developed and implemented an internal **full-stack project management platform (Next.js, ASP.NET Core, PostgreSQL)** to centralize laboratory resources and schedules, **reducing context-switching** and streamlining project tracking for the team.

[Samsung Corporation](#)

Bac Ninh, Vietnam

Software Engineer Intern

August 2023 - October 2023

- Architected and deployed a **pretrained model** computer vision system for display panel defect detection, **reducing manual inspection latency by 40%** and standardizing production **QA benchmarks**.
- Developed a **C#/.NET (WPF)** desktop application with a **SQL Server** backend to centralize resource tracking, **replacing fragmented spreadsheet workflows** and **improving cross-team operational visibility**.

[Vietnam Posts and Telecommunications Group](#)

Hanoi, Vietnam

Data Engineer Associate

October 2022 - June 2023

- Built distributed data pipelines (**PySpark, Hadoop**) processing **100M+ records**, **improving performance 2×** and **supporting downstream backend services**.
- Engineered a full-stack recommendation engine (**FastAPI, Next.js, MongoDB**) using **Selenium** pipelines to **eliminate "dead-end" user experiences** by delivering real-time alternatives for out-of-stock items, directly **increasing user retention** and **recapturing potential lost revenue**.

CERTIFICATION

AWS Certified Developer – Associate (Valid until Feb 2029) - [\[Certificate Link\]](#)

PROJECTS

Rate Pulse (December 2025 – Present) | [Github](#)

- Built** and deployed a full-stack foreign exchange platform that aggregates rates, trends, and related news from multiple sources into a centralised dashboard for currency comparison and market context.
- Developed a **Go/Gin** backend with **REST** and **gRPC** services, **Protocol Buffers**, **PostgreSQL/sqlc** persistence, **Redis** caching, and **Paseto/JWT** authentication for secure service workflows, and **Brevo** email integration.
- Built a **Next.js** dashboard with rate comparison views, trend visualisations, real-time monitoring, AI-generated market insight cards, and **Brevo-powered contact email delivery**.
- Created a **Python + Selenium** data collection pipeline to normalise structured exchange-rate and market data from multiple external sources for downstream analysis.
- Automated CI/CD** with **GitHub Actions** and **Docker** Hub, deploying containerised services through **Kubernetes** on **DigitalOcean** after migrating from **AWS** to **reduce infrastructure cost**.

Book Borrow (August 2025 – November 2025) (Team of 8) | [Github](#)

- Engineered and delivered a production-ready P2P marketplace for a **non-profit book reader organisation in Western Australia**, utilizing **FastAPI** and **Next.js** with integrated **Stripe** payments, **Google OAuth**, and **AWS RDS**.

TECHNICAL SKILLS

- Programming Languages:** Golang, Python, C/C++, JavaScript/TypeScript, Java, SQL (PostgreSQL, MySQL).
- Frameworks & Libraries:** ASP.NET Core, Next.js, React, FastAPI, Gin (Go), Flask, Qt (C++), PySpark.
- Cloud & DevOps:** DigitalOcean, AWS, Docker, Kubernetes, GitHub Actions, Vercel, Render, Supabase, Upstash.
- Databases & Caching:** MongoDB, Redis.
- Tools & Architectures:** Git, gRPC, RESTful APIs, Brevo, Selenium, Linux, Unit Testing, CI/CD Pipelines.

EDUCATION

[The University of Western Australia](#)

Perth, Australia

Master, Information Technology

February 2024 – December 2025

[Hanoi University of Science and Technology](#)

Hanoi, Vietnam

Bachelor, Information Technology

August 2019 - September 2023